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ABSTRACT

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`\beginabstract ...\endabstract` and `\begin{keyword} ... \end{keyword}` which contain the abstract and keywords respectively.
Each keyword shall be separated by a `\sep` command.

1. Introduction

2. Related works

2.1. Deep Learning-based IoT Intrusion Detection

Phalaagae, Zungeru, Yahya, Sigweni, and Rajalakshmi (2025) proposed hybrid deep learning model combining CNN and bidirectional LSTM with an attention mechanism, used Particle Swarm Optimization (PSO) for dynamic feature selection and and convolutional layers, two layers (128 and 64 filters) to extract spatial features, Max-pooling to reduce dimensionality, and two bidirectional LSTMs with two layers (64 and 32 units) to capture temporal dependencies. The attention mechanism is applied to LSTM hidden states to compute alignment scores and weights, focusing on critical time steps. Datasets: IoTID20 (primary), CICIoT2023, and Edge-IIoTset (for generalization). Metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix, Training Loss. Performance (IoTID20): Accuracy: 98.88% ,F1-Score: 99.36%, Precision: 99.91% ,Recall: 98.82%. Limitations: Although effective, deployment in real-time distributed IoT environments may still face scalability/complexity challenges. Slight reduction in performance on highly imbalanced/complex datasets (CICIoT2023). Future Work: Explore transformer architectures; implement real-time threshold optimization; test other metaheuristics like Genetic Algorithms (GA) or Ant Colony Optimization (ACO).

Fares, Ibrahim, Elaziz, Shrahili, Elmahallawy, Sohaib, Shawky, and Shah (2025) proposed Hybrid Swin Transformer + LSTM with Transfer Learning. in the input transformation layer, it converts 1D traffic data into a format suitable for Swin (Patch Partition) , Swin Transformer layer extracts hierarchical, local, and global features using shifted windows, and LSTM Layers process the feature sequence from Swin to capture long-term temporal dependencies. in Transfer Learning Strategy: Pre-training: Train on source dataset (NSL-KDD). Transfer: Move weights to target model. Fine-tuning: retrain on target IoT datasets (ToN-IoT, BoTIoT, etc.) with reduced parameters (frozen layers). Metrics: Accuracy, Precision, Recall, F1-Score, Detection Rate (DR), False Positive Rate (FPR), Processing Time.

Overall Performance: Average Accuracy 98.97%, F1 98.97%. CICIoT2023: 98.78% Binary Accuracy, 97.44% Multiclass. ToN-IoT: 99.97% Binary Accuracy. Training Time: Transfer learning reduced training time significantly (e.g., 950s for CICIoT2023 vs 2072s for autoencoders). Detection Rate: 99.14% on CICIoT2023 (Highest among compared models).

Limitations: High computational complexity of Transformers (mitigated by TL but still non-negligible). Dependency on the quality of the source dataset (NSL-KDD) for initial feature learning. Future Work: Improve interpretability; Enhance scalability

Racherla, Sripathi, Faruqui, Kabir, Whaiduzzaman, and Shah (2024) developed Deep-IDS , a lightwieght LSTM basd (input equential network traffic data, hidden layer with 64 LSTM units and memory cells/gates to capture temporal dynamics) and output softmax classification into 6 classes (Benign + 5 Attacks(Denial of Service (DoS), Distributed Denial of Service (DDoS), Brute Force (BRF), Man-in-the-Middle (MITM), Replay Attacks (RP))) , Traffic blocking based on a probability threshold. Empirically set at 70% to optimize the detection rates and false alarms. they used CIC-IDS2017 dataset and Min-Max Normalization, One-Hot Encoding, Preprocessing Mutual Information Feature Selection Metrics : Accuracy:97.67%, Precision :97.67% , Recall : 98.17%, F1-Score, FAR, Detection Rate :96.8% at the optimal 70% threshold, Response Time : 1.49 seconds on average (Real-time capable) .

2.2. Autoencoder and Generative Intrusion Detection

Wei, Yin, Tan, Ruan, Yin, Sun, and Luo (2024) developed Dual-Classifier Hybrid Model (AHDM) , used Autoencoder Feature Enhancement (AFE) for Feature Grouping (Divides features into 4 subgroups based on statistical metrics (mean, std, skewness)) and Feature Encoding (4 distinct Autoencoders encode these subgroups into lower-dimensional spaces ($k_g = 4 - g$) to highlight subtle differences.).

Classifier I (Raw Features : DNN trained on original features to detect normal traffic and large-sample attacks). Classifier II (Encoded Features : DNN trained on AFE-enhanced features specifically to detect small-sample attacks). Decision Fusion: Combines outputs from both classifiers.

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Metrics: Accuracy, Precision, Recall, F1-Score, False Alarm Rate (FAR). Datasets : KDD CUP 99, CIC-IDS-2017, IoT-23.

KDD CUP 99: 99.76% Accuracy, 0.06% FAR.

Yu, Zhai, Shen, Wu, Zhang, Yu, and Shen (2024) combined Deep Q-Networks (DQN) with Conditional Variational Autoencoders (CVAE) , Known Traffic Classifier (DQN Module): Agent(DQN Agent), Values Network(A CVAE structure integrated into the Q-network) , Learns to classify known classes (Benign + Known Attacks).

Unknown Attack Recognizer (Reconstruction Module): Takes latent features (z) from the trained Recognition Network and predicted labels (a^*) to reconstruct features , Uses Reconstruction Error + Extreme Value Theory (EVT) threshold. High error → Unknown Attack.

Double Deep Q-Network (DDQN) : Used for the classification task to mitigate Q-value overestimation. Conditional Variational Autoencoder (CVAE): Integrated to learn latent distributions of known classes.

Extreme Value Theory (EVT) : Models the tail of the reconstruction error distribution (Generalized Pareto Distribution) to mathematically define the "unknown" threshold.

Known (Training): DDoS, Password, XSS. Unknown (Testing): DoS, Injection, Scanning, MITM. (TON-IoT Dataset). Unknown Detection : Recognized 82% of unknown attacks (which were otherwise misclassified as DDoS/Password by the classifier)

Explicit Limitations: Computational complexity depends on DQN training episodes/steps ($Q(m \times n)$). Still some misclassification (18% of unknown attacks not recognized). Future Work: Adapt to constantly emerging cyber attacks; new DRL algorithms for open-set.

2.3. Reinforcement Learning-based Intrusion Detection

Rookard and Khojandi (2023) applied Deep Q-Network (DQN) device-specific telemetry data (GPS, Modbus) from the TONIoT dataset, Environment: IoT Network Traffic (TON-IoT samples). State : input features from IoT telemetry (e.g., temperature, pressure, location). Agent: Deep Neural Network approximating the Q-function. Action: Classification decision (Attack vs Normal). Reward : Feedback based on correct/incorrect classification.

Attacks Considered: Confidentiality: Scanning, Password cracking, SQL Injection. Integrity: Backdoor, XSS. Availability : Ransomware, DDoS.

Dataset: TON-IoT (Telemetry from GPS Tracker, Modbus, Weather Station).

Metric: Accuracy, Precision, Recall, F1-Score, Geometric Mean (G-Mean).

Results: GPS Dataset: DQN Accuracy 79.69% (Highest). G-Mean 0.8854. Modbus Dataset : DQN Accuracy 80.51% (Significantly outperformed MLP at 69% and SVM at 68%). Weather Dataset: DQN Accuracy 87.82%(Highest). DQN consistently outperformed supervised learning baselines across all three device types.

Future Work : Explore Actor-Critic methods; feature engineering; testing on additional RL algorithms.

Ghubaish, Yang, and Jain (2024) proposed HDRL-IDS, utilizing a Deep Deterministic Policy Gradient (DDPG) agent for hybrid hostnetwork intrusion detection in 5G medical applications.

Need for a Hybrid IDS (combining NIDS and HIDS) using DRL (Actor-Critic) that adapts to 5G MEC environments with minimal human interaction, validated on a realistic 5G testbed.

Environment: 5G MEC Testbed (Simu5G + OMNet++). Agent: DDPG (Deep Deterministic Policy Gradient) Agent. State Space: Hybrid features combining Network metadata (via Argus) and Host metadata (via Psutil). Action Space: Continuous control (mapped to classification). Reward Function: Feedback on detection accuracy.

Testbed: Includes 5G Core, Local Network, 2 MECs, 3 UEs, Insider Attacker.

Algorithms Used: Deep Deterministic Policy Gradient (DDPG): Model-free, off-policy Actor-Critic algorithm suitable for continuous action spaces.

Actor Network: 4 Layers (32 GRU, 48 Dense, 48 Dense, 1 Output).

Critic Network : Evaluates the actor's policy. Strategy: Epsilon-greedy for exploration.

MITM : Data alteration between UE and MEC. DDoS : Flooding MECs. Ransomware: Encrypting files on UE. Buffer Overflow: Manipulating execution flow.

Dataset: Novel self-generated 5G MEC dataset (145k samples: 132k normal, 13k attack). 77 features (35 network, 40 host). Hybrid (HDRL): Outperforms single-feature models in F1-score. Net-only: High accuracy (91.76%) but failed to identify attacks (0% detection of minority class), leading to low F1 (10.48%). Host-only: Low accuracy (24.99%) and low F1 (4.45%). Hybrid: Accuracy 58.17%, F1 25.42%. (Note: Results are initial and low in absolute terms, but Hybrid is relatively superior in balanced detection).

Limitations: Initial results show relatively low absolute performance (F1 25%, Accuracy 58%). Predefined parameters might not be optimal. Low number of training episodes (10) limited the model's potential.

Future Work: Increase training episodes; Feature reduction methods; Deployment optimization.

Jin, Fan, Han, and Cui (2025) combined Soft ActorCritic (SAC) with an Improved Gravitational Search Algorithm (IGSA) for robust feature selection. Improved GSA uses fitness normalization with Adaptive Search Radius and a sigmoid-modulated gravitational constant to select optimal features. Intrusion Detection Module (SAC): Agent: SAC Agent (Actor-Critic). Input: Selected features from IGSA. Networks: Actor Network, V-Critic Network, Q-Critic Network (all 3-layer DNNs with 100 neurons each).

Algorithms Used:

Improved Gravitational Search Algorithm (IGSA):

Fitness Normalization : Uses median/quartiles for robustness against outliers ($m_i(t)$ calculation).

Adaptive Search Radius: Dynamic adjustment ($R(t)$) to balance exploration/exploitation.

Sigmoid Modulation: Modulates gravitational constant $G(t)$ to prevent premature convergence.

Soft Actor-Critic (SAC): Off-policy DRL maximizing expected cumulative reward + entropy (randomness) to prevent deterministic overfitting. Uses Twin Q-functions to mitigate overestimation bias.

Attacks : DoS, Probe, R2L, U2R (NSL-KDD); Injection, Flooding (AWID).

Datasets: NSL-KDD, AWID, CICIDS2017.

Metrics: Accuracy, Precision, Recall, F1-Score, Training/Inference Time.

Results: NSL-KDD: Accuracy 84.15%, F1 84.85%. Outperformed GA, PSO, and other DRL baselines (AE-RL, SHIA, SSDDQN). AWID: Accuracy 98.9%, F1 > 98.9%. CICIDS2017: Accuracy 98.41%. Efficiency: Feature reduction (122 → 50 for NSL-KDD) reduced training time by 33% (180s → 120s compared to GA-SAC).

Limitations: Struggles with U2R (User-to-Root) category due to limited training samples (class imbalance).

Future Work: Explore SMOTE and GANs for data augmentation to address U2R imbalance.

Severin, Eric, Moussa, Vianey, Armand, and Pierre (2025) employed AE-LSTM (Feature Extraction) and Centralized MARL (Decision Making)

AE-LSTM (Autoencoder LSTM):

Encoder: 3-layer LSTM (64 units) to compress time-series network flows into latent vectors (z).

Decoder: 3-layer LSTM to reconstruct the input. MSE loss used for training.

Multi-Agent Environment (MultiIoT23Env):

Agents: Multiple PPO Agents (5 agents mentioned: Agents 0-4).

Structure : Centralized Training, Decentralized Execution (CTDE) paradigm

Input: Latent vectors from AE-LSTM.

Action: Binary classification (0: Normal, 1: Attack).

Dataset: IoT-23 (Subset: 5% of original, split into 80% train / 20% test). Preprocessing: Cleaning, Correlation removal (>0.95), Normalization (StandardScaler), and Class Balancing (SMOTE-ENN).

Metrics: Precision:99.30%, Recall: 99.35%., F1-Score: 99.40 %.

Limitations: Complexity of multi-agent configuration relative to single-agent. Resource constraints on real IoT devices (CPU/Memory) need validation.

Future Work: Integration of attention mechanisms (Transformers); Validation on real-world resource-constrained hardware; Dynamic agent policy management.

2.4. Research Motivation

Despite the notable succes of recnet Deep Reinforcement Learning (DRL) and generative models in IoT Intrusion detection , there is two critical gaps unaddressed . First , existing models heavily rely on external data augmentation techniques such as GANs and SMOTE to mitigate extreme class imbalance such as the U2R class in Jin et al. (2025) , which artificially alters the original data distribution and increases computational overhead. Second , when mitigating zero days attacks , current methos primarily rely on static anomaly detection thresholds such as EVT in Yu et al. (2024) that flag novel attacks as generic unknowns without dynamically learning their signatures, often resulting in high miss rates.

To address these critical limitations, we proposes a novel framework: Continual Open-Set Reinforcement Learning with Rarity-Aware Dynamic Reward Shaping. Handling imbalance via reward shaping, instead of preprocessing or generating synthetic data, the reward function dynamically scales the agent's rewards to the class probability , this natively forces the RL agent to prioritize minority classes without altering the dataset.

Continual zero-day attack discovery, rather than statically flagging unknown traffic, the RL agent clustering and learning new zero-day attacks as distinct classes without forgetting previously known attacks.

By bridging dynamic reward shaping with continual open-set learning, this framework aims to provide a highly adaptive and suitable solution for dynamic IoT environments.

3. Problem Formulation

In dynamic IoT environments, network traffic is continuously generated by the heterogenous devices with highly imbalanced attacks and the potential emergence of previously unseen attacks (zero-day). We formulate the IoT intrusion detection task as continual open set reinforcement learning problem over an imbalanced data stream.

3.1. IoT Traffic Representation

Let IoT Traffic stream be represented as a sequence of feature vectors :

$$x_t \in \mathbb{R}^d \tag{1}$$

Where d denotes the number of extracted traffic features after after preprocessing and normalization.

Each traffic instance is associated with a class label :

$$y_t \in C = \{c_1, c_2, ..., c_K\} \tag{2}$$

Where C represents the set of known attack categories and benign traffic.

Due to real-world imbalance, the empirical class distribution is defined as :

$$p(c_i) = \frac{N_{c_i}}{N_{\text{total}}} \tag{3}$$

where N_{c_i} denotes the number of samples belonging to class c_i , and N_{total} is the total number of samples.

In practical IoT datasets , certain class satisfy :

$$p(c_i) \ll p(c_j), \quad \exists i, j \tag{4}$$

indicating severe minority class imbalance.

MDP Formulation We model intrusion detection as a Markov Decision Process (MDP) defined as :

$$\mathcal{M} = (S, A, P, R, \gamma) \tag{5}$$

where:

- S is the state space, defined as $S = \mathbb{R}^d$, representing normalized IoT feature vectors.
- A is the action space corresponding to classification decisions:

$$A = \{1, 2, ..., K\} \tag{6}$$

- $P(s_{t+1}|s_t, a_t)$ denotes the environment transition probability.
- $R(s_t, a_t)$ is the reward function.
- $\gamma \in (0, 1]$ is the discount factor.

At each timestep t, the agent observes state $s_t = x_t$, selects an action $a_t \in A$, and receives a reward R_t based on prediction correctness.

The objective is to learn an optimal stochastic policy $\pi_\theta(a|s)$ that maximizes the expected discounted cumulative reward:

$$\pi^* = \arg\max_{\pi} \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R_t \right] \tag{7}$$

3.2. Rarity-Aware Reward Definition

To address class imbalance without altering the original data distribution, we introduce a rarity-aware reward shaping mechanism. Let R_{base} denote the base classification reward:

$$R_{\text{base}} = \begin{cases} +1, & \text{if } a_t = y_t \\ -1, & \text{otherwise} \end{cases} \tag{8}$$

We define the rarity-aware reward as:

$$R_t = R_{\text{base}} \cdot \left(1 + \alpha \log \left(1 + \frac{1}{p(y_t)} \right) \right) \tag{9}$$

where $\alpha > 0$ controls the strength of imbalance compensation. This formulation increases the reward contribution of minority classes while preserving the original dataset distribution.

3.3. Open-Set Intrusion Detection

In dynamic IoT networks, previously unseen attacks may appear during deployment . Let x_u denote a traffic sample drawn from an unknown distribution:

$$x_u \sim P_{\text{unknown}}, \quad x_u \notin \mathcal{C} \tag{10}$$

The agent must satisfy the following requirements:

1. Detect high-uncertainty samples that do not belong to known classes.
2. Assign them to a temporary unknown buffer.
3. Expand the class set:

$$\mathcal{C} \leftarrow \mathcal{C} \cup \{c_{K+1}\} \tag{11}$$

4. Update the policy π_θ without catastrophic forgetting.

3.4. Continual Learning Objective

Let $\mathcal{C}_t$ denote the set of known classes at time t. The learning objective extends to:

$$\max_{\pi_\theta} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R_t \right] \quad \text{subject to} \quad \mathcal{C}_t \subseteq \mathcal{C}_{t+1} \tag{12}$$

while minimizing performance degradation on previously learned classes.

This formulation defines intrusion detection as a continual open-set reinforcement learning problem under severe class imbalance, forming the theoretical foundation of the proposed CORL-IDS framework.

4. Proposed CORL-IDS Framework

This section presents the proposed Continual Open-Set Reinforcement Learning Intrusion Detection System (CORL-IDS). The framework integrates rarity-aware reward shaping, entropy-based open-set detection, and continual class expansion into a unified reinforcement learning architecture.

4.1. Overall Architecture

The proposed framework consists of four main components :

1. Feature Encoding Module
2. Rarity-Aware Soft Actor-Critic Agent
3. Entropy-Based Novelty Detection Module
4. Continual Class Expansion with Stability Regularization

At each timestep t , an incoming IoT traffic instance x_t is encoded into a latent representation z_t , processed by the RL agent for classification, and evaluated by a novelty detection mechanism. If a sample is identified as unknown , it is temporarily stored and later used for class expansion.

4.2. Feature Encoding Module

IoT traffic exhibits temporal dependencies across sequential flows.

To capture temporal dynamics while maintaining computational efficiency, we employ a lightweight LSTM encoder.

Let $x_t \in \mathbb{R}^d$ denote the normalized feature vector. The encoder maps x_t into a latent representation:

$$z_t = f_\phi(x_t) \tag{13}$$

where $f_\phi(x_t)$ represents a single-layer LSTM with 64 hidden units followed by a linear projection layer.

The latent vector $z_t \in \mathbb{R}^h$ serves as the state representation for the reinforcement learning agent.

4.3. Rarity-Aware Soft Actor-Critic Optimization

We adopt the soft actor-critic (SAC) framework as the core learning algorithm due to its stability and entropy-regularized exploration.

The stochastic policy is defined as

$$\pi_\theta(a|z) \tag{14}$$

SAC optimizes the entropy-regularized objective:

$$J(\pi) = \mathbb{E}_{(z_t,a_t)\sim\pi}\left[\sum_{t=0}^{\infty}\gamma^t\left(R_t + \alpha_{\text{ent}}H(\pi(\cdot|z_t))\right)\right] \tag{15}$$

where $H(\cdot)$ denotes policy entropy and α_{ent} controls exploration.

4.3.1. Rarity-Aware Reward Shaping

To address severe class imbalance without altering the original data distribution , we introduce a log-scaled rarity-aware reward:

$$R_t = R_{\text{base}}\left(1 + \lambda\log\left(1 + \frac{1}{p(y_t)}\right)\right) \tag{16}$$

where:

- $R_{\text{base}} \in \{+1, -1\}$,
- $p(y_t)$ is the empirical class probability,
- λ controls imbalance compensation strength.

This formulation amplifies minority-class rewards while preventing instability via logarithmic scaling.

4.3.2. Critic Update

Twin Q-networks $Q\omega_1$ and $Q\omega_2$ are optimized using the Bellman residual:

$$J_Q(\omega_i) = \mathbb{E}\left[\left(Q_{\omega_i}(z_t, a_t) - y_t\right)^2\right] \tag{17}$$

where the target value is :

$$y_t = R_t + \gamma\left(\min_{j=1,2} Q^{\text{target}}_{\omega_j}(z_{t+1}, a_{t+1}) - \alpha_{\text{ent}}\log\pi_\theta(a_{t+1}|z_{t+1})\right) \tag{18}$$

4.4. Entropy-Based Novelty Detection

To enable open-set intrusion detection , we measure policy uncertainty using entroy:

$$H_t = -\sum_{a\in A}\pi_\theta(a|z_t)\log\pi_\theta(a|z_t) \tag{19}$$

An adaptive threshold is defined as:

$$\tau_t = \mu_H + \beta\sigma_H \tag{20}$$

where μ_H and σ_H denote running mean and standard deviation of entropy values and β control sensitivity.

If :

$$H_t > \tau_t \tag{21}$$

the sample is considered potentially unknown and stored in an unknown buffer $\mathcal{B}_u$ for further analysis.

4.5. Continual Class Expansion

Unknown samples accumulated in $\mathcal{B}_u$ are periodically clustered using density-based clustering If a stable cluster $\mathcal{G}$ is identified with density, a new class c_{K+1} is created

$$\mathcal{C} \leftarrow \mathcal{C} \cup \{c_{K+1}\} \tag{22}$$

4.5.1. Stability Regularization

To prevent catastrophic forgetting, we incorporate Elastic Weight Consolidation (EWC).

The regularized loss becomes

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SAC}} + \frac{\eta}{2}\sum_i F_i(\theta_i - \theta_i^*)^2 \tag{23}$$

where:

- F_i is the Fisher Information matrix diagonal,
- θ_i^* denotes previously learned parameters,
- η controls stability strength.

4.6. Learning Procedure

The complete training process alternates between:

1. Rarity-aware SAC optimization
2. Entropy-based unknown detection
3. Periodic clustering and class expansion
4. Continual policy update with EWC

This unified design enables adaptive minority-class reinforcement, dynamic zero-day discovery, and continual knowledge retention within a single reinforcement learning framework.

5. Experimental Setup

This section describes the datasets, preprocessing pipeline, evaluation metrics, baseline models, and experimental protocols used to validate the proposed CORL-IDS framework.

5.1. Datasets

To evaluate robustness under heterogeneous IoT conditions , experiments are conducted on two publicly available intrusion detection datasets:

- **CICIDS2017** : A widely used benchmark containing benign and multiple attack categories including DoS, DDoS, Brute Force, Infiltration, Botnet, and Web attacks.
- **TON-IoT** : A realistic IoT telemetry dataset containing network and device-level attacks such as Scanning, DDoS, Ransomware, Injection, and Backdoor.

These datasets were selected due to :

1. Severe class imbalance in minority attack categories.
2. Presence of diverse attack families suitable for open-set evaluation.
3. Realistic IoT network traffic characteristics.

5.2. Data preprocessing

The following preprocessing steps are applied :

- Removal of duplicate and corrupted entries.
- Missing value imputation.
- Min-Max normalization applied to all numerical features.
- Temporal ordering of traffic flows for sequential modeling.

No data generation was applied, such as SMOTE or GAN-based augmentation, to preserve the natural class distribution.

5.3. Imbalance Configration

To evaluate robustness under extreme imbalance, minority attack classes with probability:

$$p(c_i) < 0.02 \tag{24}$$

are explicitly retained without resampling.

The proposed rarity-aware reward shaping is responsible for imbalance mitigation.

5.4. Open-Set Experimental Protocol

To simulate zero-day attack conditions, we adopt the following open-set protocol:

1. Select a subset of attack classes as known classes during training.
2. Remove one or more attack categories from the training set.
3. Introduce the removed classes only during testing.

Let:

$$\mathcal{C}_{train} \subset \mathcal{C}_{test} \tag{25}$$

Unknown classes are treated as zero-day attacks and evaluated using the entropy-based novelty detection mechanism.

5.5. Continual Learning Scenario

To simulate dynamic IoT deployment, the data stream is divided into sequential phases:

- Phase 1: Training on initial known classes.
- Phase 2: Introduction of unknown attack samples.
- Phase 3: Class expansion and policy adaptation.

5.6. Baseline Models

The proposed CORL-IDS framework is compared against :

- Vanilla Soft Actor-Critic (without reward shaping).
- Soft Actor-Critic with SMOTE-based oversampling.
- Deep Q-Network (DQN).
- Proximal Policy Optimization (PPO).

All baseline models use identical feature inputs for fairness.

5.7. Evaluation Metrics

To provide a comprehensive evaluation under imbalance and open-set conditions, the following metrics are reported:

- Accuracy
- Precision
- Recall
- F1-score (Macro and Minority-class specific)
- Balanced Accuracy
- False Positive Rate (FPR)
- Open-set Detection Rate (Unknown Recall)
- Catastrophic Forgetting Measure:

$$F = \frac{1}{K} \sum_{i=1}^{K} \left(\max_t R_{i,t} - R_{i,T} \right) \tag{26}$$

where $R_{i,t}$ denotes performance on class i at phase t, and T is the final phase.

5.8. Implementation Details

The model is implemented in PyTorch.

Key hyperparameters:

- Discount factor $\gamma = 0.99$
- Entropy coefficient α_{ent} automatically tuned
- Rarity scaling coefficient $\lambda \in \{0.5, 1.0, 2.0\}$
- LSTM hidden units: 64
- Replay buffer size: 100,000
- Batch size: 256
- Optimizer: Adam with learning rate 3×10^{-4}

All experiments are repeated over three independent runs, and average performance is reported.

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